MIL-A-2402D
1 November 1978
SUPERSEDING
MIL-A-2402C
4 January 1971

MILITARY SPECIFICATION

ADAPTER, STEREOPLOTTER PROJECTOR, TILT, WITH CONDENSER CLIP (FOR MULTIPLEX)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tilt adapter and a condenser clip for wide-angle multiplex projectors.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13217E9306	- Adapter, Stereoplotter.
TA13217E7485	- Clip, Condenser.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The tilt adapter with condenser clip shall be in accordance with Top Assemblies TA13217E9306 and TA13217E7485, respectively, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the tilt adapter with condenser clip. Any data (e.g. shop drawings, layouts,

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flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first produced tilt adapter with condenser clip). The contractor shall furnish one or more tilt adapters with condenser clips as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce tilt adapters with condenser clips that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.4 Identification marking. The tilt adapter with condenser clip shall be identified in accordance with MIL-STD-130.

3.5 Treatment and painting. The portions of the tilt adapter with condenser clip normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color and gloss as specified (see 6.2).

3.6 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.7 Workmanship. All parts, components, and assemblies of the tilt adapter with condenser clip including castings, molded parts, stampings, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges and corners are required.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsibile for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of any examination by the first produced tilt adapter with condenser clip shall be cause for disassembly, in the presence of a Government representative, of the first produced tilt adapter with condenser clip to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced tilt adapter with condenser clip. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced tilt adapter with condenser clip inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

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4.3 First produced tilt adapter with condenser clip inspection.

4.3.1 Examination. The first produced tilt adapter with condenser clip shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.1.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The tilt adapter with condenser clip shall be examined as specified herein for the following defects:

Major

101. Dimensions not in accordance with drawings.
102. Material not as specified on the drawings.
103. Parts not interchangeable.
104. Workmanship not as specified.
105. Castings not as specified on the drawings.
106. Brazing not smooth (condenser clip only).

Minor

201. Identification marking not as specified.
202. Treatment and painting not as specified.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials or containers not as specified for Level A or Level B. Each incorrect material or container shall constitute one defect.
108.	Boxes not closed Method V as specified.
109.	Strapping not as specified.
110.	Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each tilt adapter with condenser clip shall be preserved in accordance with MIL-P-116, Method III, using a close-fitting box conforming to PPP-B-636, W6c, style optional as the container. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. Tilt adapters with condenser clips shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Tilt adapters with condenser clips, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional, or PPP-B-621, Class 2, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Tilt adapters with condenser clips, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 350 or PPP-B-640,

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Class 1, style optional. The boxes shall be closed in accordance with the appendix to the applicable box specification. Strapping shall be Finish A.

5.2.3 Commercial. Tilt adapters with condenser clips shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The tilt adapter is intended to tilt a vertical multiplex projector so that it may be used with oblique photographs. The condenser clip is intended to securely anchor the condenser system when the projector is in a tilted position.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced tilt adapter with condenser clip and number of tilt adapters with condenser clips required (see 3.2).
(c) Color and gloss required (see 3.5).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When Finish A strapping is required (see 5.2.1).

6.3 First produced item. Any changes or deviations of production tilt adapters with condenser clips from the approved first produced item during production will be subject to the approval of the contracting officer. Approval of the first produced item will not relieve the contractor of his obligation to furnish tilt adapters with condenser clips conforming to this specification.

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6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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